Grade 4 Mathematics: Whole numbers, fractions and perimeter

Mathematics · Grade 4 · 29 marks · 60 minutes

Educator marking memo

Suggested answers and marking points. Accept equivalent correct responses appropriate to the grade.

Assessment

Read the stimulus and each question carefully.

> *A school garden has 8 rows with 24 seedlings in each row. The garden is a rectangle 8 m long and 5 m wide. Learners water three quarters of the garden on Monday.*

1.1 (5 marks) Calculate the total number of seedlings. Show a method.

- Recognises 8 equal groups of 24. (1 mark)
- Uses multiplication or repeated addition. (1 mark)
- Calculates 8 x 20 = 160 or equivalent. (1 mark)
- Calculates 8 x 4 = 32 or equivalent. (1 mark)
- Adds to give 192 seedlings. (1 mark)

1.2 (5 marks) Learners measured the garden’s four sides with a tape: 8 m, 5 m, 8 m and 5 m. Add the four measurements to find the perimeter.

- Identifies the four measured side lengths as 8 m, 5 m, 8 m and 5 m. (1 mark)
- Chooses to add all four side measurements to find the perimeter. (1 mark)
- Adds the two 8 m sides to get 16 m. (1 mark)
- Adds the two 5 m sides to get 10 m. (1 mark)
- Adds the side totals to get 26 m and includes the unit. (1 mark)

1.3 (6 marks) What fraction of the 192 seedlings is 48? Simplify it. Show how you simplify the fraction and check your answer.

- Writes the fraction 48/192. (1 mark)
- Identifies 48 as a common divisor of 48 and 192, or shows another valid common-divisor method. (1 mark)
- Shows the numerator divided by the chosen common divisor, for example 48 ÷ 48 = 1 or 48 ÷ 16 = 3. (1 mark)
- Shows the denominator divided by the same common divisor, for example 192 ÷ 48 = 4 or 192 ÷ 16 = 12. (1 mark)
- Gives the simplified fraction 1/4. (1 mark)
- Checks the result by showing that 1/4 of 192 is 48. (1 mark)

1.4 (7 marks) The class has 360 seeds and plants 145. Show the column subtraction, make any regrouping clear, and check your answer with addition. How many seeds remain?

- Chooses subtraction and writes 360 - 145. (1 mark)
- Shows the regrouping needed to subtract the ones correctly. (1 mark)
- Subtracts the ones correctly. (1 mark)

- Subtracts the tens correctly. (1 mark)
- Subtracts the hundreds correctly. (1 mark)
- Gives 215 seeds remaining. (1 mark)
- Checks the result with addition, for example 215 + 145 = 360. (1 mark)

1.5 (6 marks) Each seedling needs 250 mL of water. For 8 seedlings, show the multiplication and the conversion calculation. Give the total in millilitres and litres.

- Sets up 8 groups of 250 mL, for example 250 × 8. (1 mark)
- Calculates 2,000 mL. (1 mark)
- Uses the relationship 1,000 mL = 1 L. (1 mark)
- Shows division by 1,000 or an equivalent conversion method. (1 mark)
- Gives the converted answer as 2 L. (1 mark)
- Includes the correct mL and L units with the two totals. (1 mark)

Marking checks

Confirm section totals equal the stated total. Apply the grade-appropriate memo style in the Grade Profile and moderate open-ended answers before use.

Source basis: DBE CAPS Mathematics · intermediate_grade_4_6. See the grade receipt for the official PDF URL and SHA-256.

DRAFT — Qualified educator review required before classroom use.